

WASP-50b

R-0714

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_241019 · created 2026-07-27T23:12:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460600.84 +/- 0.11 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.023 +/- 0.051
Transit depth (Rp/Rs)^2	0.054 +/- 0.24 [%]
Orbital Inclination (inc)	80.61 +/- 2.87
Airmass coefficient 1 (a1)	0.8803081 +/- 1.2e-06
Airmass coefficient 2 (a2)	0.04 +/- 0.027
Scatter in the residuals of the lightcurve fit is	5.8e-05 %
Best Comparison Star	#3 - [88, 430]
Optimal Aperture	31.19
Optimal Annulus	118.53
Transit Duration (day)	0.0003 +/- 0.0015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.023 $\pm$ 0.051 vs 0.1404 (deviation 2.3 σ)
Duration fit vs published	0.0003 d vs 0.0754 d (deviation 50.08 σ)
Mid-time O-C	-2753.8 min
Depth SNR	0.5
Residual scatter	5.8 %

Light curve

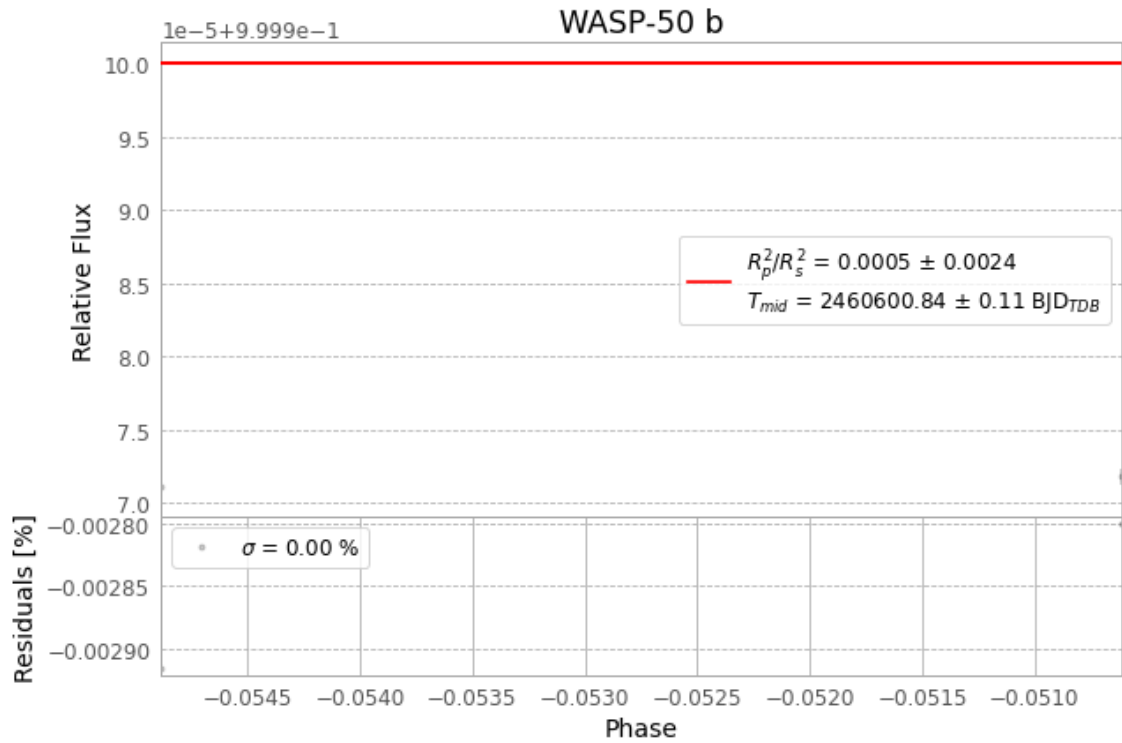


Figure 1 — FinalLightCurve_WASP-50 b_2024-10-18.png (SHA-256 91abfc706abb0469...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [116, 471], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 811b60dfead8a92a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8a6a059

Data provenance

77 FITS frames (51 MB), WASP-50241019040315.FITS → WASP-50241019075116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-241019.log (141df343da59...), FinalLightCurve_WASP-50 b_2024-10-18.png (91abfc706abb...), FinalLightCurve_WASP-50 b_2024-10-18.csv (4b4b8a537aaa...)

Compute resources

Wall time 765.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 23:12Z.